



DEVELOPERS LEVEL · SEMESTER 2

Healthy Snack Calculator

Student Workbook

Code takes input, does the math, and shows output — to support real choices.

Name: _____

THEORY

Code Can Calculate

Many tools follow one pattern: take **input**, do a **calculation** with **operators** (+ - * /), and show **output**. **Variables** hold the numbers along the way. A snack calculator takes servings and calories-per-serving as input and outputs the total.

Discuss: what inputs would a healthy-snack calculator need, and what output would help you choose?

Trace it

Read the code and give the output.

```
let perServing = 90;
```

```
let servings = 3;
```

```
let total = perServing * servings;
```

What is the output (total)? _____

Code we use:

input

+ - * /

let (variable)

function

output

console.log

Words we are learning

- **Input** — information the user gives the program.
- **Output** — the result the program shows back.
- **Operator** — + - * / for add, subtract, multiply, divide.
- **Variable** — stores each number the program uses.
- **Calculation** — using operators to turn input into output.

Name: _____ Date: _____

Math Operators

Match each **operator** to what it does, then solve.

+	add
-	subtract
*	multiply
/	divide

Solve: $90 * 3 = \underline{\quad}$ $120 / 4 = \underline{\quad}$ $50 + 35 = \underline{\quad}$

Name: _____ Date: _____

Input In, Output Out

Every calculator has **input** and **output**. Label each part below.

```
let cups = 2; // ____ (input or output?)
```

```
let mlTotal = cups * 250; // the calculation
```

```
show(mlTotal); // ____ (input or output?)
```

Write which is input and which is output:

Name: _____ Date: _____

Build A Formula

Turn a real question into a calculation. Fill in the formula.

Total calories = calories per serving ____ number of servings

```
let total = perServing ____ servings;
```

Write a formula for calories left in a daily goal:

```
let left = goal ____ total;
```

Name: _____ **Date:** _____

Plan My Snack Calculator

Plan the inputs, the formula, and the output message.

Inputs I will ask for:

My formula:

let total = _____;

Output message the user sees:

sketch the calculator

Name: _____ Date: _____

PROJECT

Healthy Snack Calculator — Plan

Build an input-driven tool that helps with a real nutrition choice: take input, calculate with operators, show a helpful output.

1. What choice does my calculator help with?

2. Inputs (variables):

let _____ = _____;

let _____ = _____;

3. Calculation:

let result = _____;

4. Output the user sees:
